MoFlo

SMB Knowledge Activity Instructions

Introduction

Congratulations on advancing to the next stage of the interview process. We're excited to see how you will approach building. This assignment will test your ability to understand system architecture, follow pre-existing practices, and create meaningful functionality that aligns with the mission of MoFlo.

About MoFlo Cloud

MoFlo Cloud is a comprehensive business automation platform that provides AI-powered tools for various business functions. All of our customers are small businesses that are very much non-technical and are not great with complex interfaces or learning new technologies. We aim to do as much for them as possible rather than making them do everything within our platform.

Applications are the core products inside MoFlo Cloud. **Each app has a specific function** (e.g., social posts, emails, blogs), **and a specific connection point** (to external platforms to push content, integrate accounts, track results, etc.)

Our platform currently includes (with some examples of function + connection):

- MoSocial
    - Function: Social media content generation (captions, hashtags, images)
    - Connection: X, LinkedIn, Facebook, Instagram, Youtube, TikTok
- MoMail
    - Function: Email content generation (outbound emails, newsletters, etc.)
    - Connection: Mail Chimp, Constant Contact, etc.
- MoBlogs
    - Function: Blog content generation (blogs, case studies, articles, etc.)

    - Connection: Wordpress, Wix, Framer,, etc.

All of our apps rely on a company knowledge base that we build for each business when they sign up for MoFlo Cloud. This knowledge base is the foundation that allows every application to generate tailored, on-brand content. The quality of the knowledge base directly determines the quality of everything MoFlo produces.

Your Assignment: The Knowledge Builder

You will build your own version of our **Knowledge**, a web-based tool that scrapes a company's website and transforms the extracted data into a structured, comprehensive knowledge base.

This is one of the most critical pieces of the MoFlo platform. When a new customer signs up, we need to understand their business deeply and quickly. Your challenge is to build the system that makes that happen.

What You're Building

A functional web application where a user can:

1. **Input a company website URL**
2. **Scrape that website** to extract meaningful business information
3. **Display the scraped results** in a structured, reviewable format
4. **Allow the user to review, edit, and enhance** the extracted knowledge
5. **Save the final knowledge base** as a structured JSON output

The Knowledge Base Standard

Attached to this assignment is a PDF (`Knowledge_Outputs.pdf`) containing example knowledge base profiles for several real companies. These examples represent the **minimum standard** for the type of information we currently extract. Your scraper and output structure should cover at least the following categories:

Company Foundation

- Overview / company description

- Website, industry, business model, company role
- Year founded, legal entity type, employee count
- Main address, other locations, service locations
- Alternative company names

Positioning

- Company pitch (a compelling summary of why a customer should choose them)
- Founding story (if available)

Market & Customers

- Target buyers
- Customer needs
- Ideal customer persona
- Industry groupings
- Industry outlook (if available)
- Channels, funnels, CTAs
- Suppliers / partners

Branding & Style

- Writing style description (tone, voice, language patterns)
- Art style description
- Fonts
- Brand colors (hex values)
- Logos (URLs)

Online Presence

- Social media links (LinkedIn, Facebook, Instagram, Twitter/X, etc.)

Key People

- Names, titles, roles, gender
- Brief descriptions of their background or relevance

Offerings

- Products and/or services
- Features of each offering
- Pricing information (if available)
- Categorization of offering types

Think Bigger

The categories above are our current baseline. A major part of this challenge is demonstrating that you can **think beyond the baseline**.

Consider: if MoSocial needs to write social posts, MoMail needs to write emails, and MoBlogs needs to write articles, what additional information would make those outputs better? What else could you extract from a website that would be valuable across all of our applications?

Some directions to consider (these are prompts, not requirements):

- Competitor information or market positioning signals
- Content themes or topics the company frequently discusses
- Testimonials, reviews, or social proof
- Certifications, awards, or trust signals
- FAQ content or common customer questions
- Calls-to-action patterns and conversion language
- Unique selling propositions or differentiators
- Seasonal or time-sensitive messaging patterns
- Legal or compliance language (privacy policies, disclaimers)
- Media mentions or press coverage
- Community involvement or values-driven messaging

The best submissions will demonstrate original thinking about what makes a knowledge base truly useful for AI-powered content generation.

Technical Requirements

Architecture & Tech Stack

- **Framework:** Next.js 15 with App Router

- **Language:** TypeScript
- **Styling:** Tailwind CSS
- **State Management:** React hooks and context
- **UI Components:** Clean, professional interface (visually match our platform where possible)

Required Features

Scrape & Build Page (`/knowledge`)

This is your main page. It should include:

- **URL input** where the user enters a company website
- **Scrape trigger** that initiates the website scraping process
- A **loading/progress state** while scraping is in progress
- **Structured display of results** organized by knowledge base categories
- The ability for the user to **review and edit** the extracted information before saving
- A **save button** that converts the final knowledge base into a JSON structure
- Form validation and error handling

Important notes on scraping:

- You should build a real scraper that fetches and parses website content. This is a core part of the challenge.
- You do NOT need to call any LLM APIs. If you want to demonstrate how AI would enhance the extracted data (e.g., generating a company pitch from raw content), use placeholder/mock outputs and clearly label them.
- Focus on extracting as much structured information as possible from the raw HTML, metadata, and page content.

View/Manage Page (`/knowledge/view`)

This page shows knowledge bases that have been created. It should include:

- Display of saved knowledge base entries (from your JSON output)
- Multiple view modes (e.g., card view, table view, detailed view) that work well with the content structure

- Filtering and/or search functionality
- Edit and delete functionality
- Any additional utilities that make sense for managing knowledge base records

Additional Requirements

- Responsive design
- Error handling throughout
- TypeScript with proper type definitions
- Clean file organization (components, utilities, types, etc.)
- A `.txt` or `.json` file containing the JSON output from your scrape/build page (at least one complete example)
- A database schema design for storing the knowledge base data (documented in your README or as a separate file)

Prompting & AI Thinking (Evaluated)

Even though you are not required to integrate a live LLM, we are evaluating your **prompting skills and AI thinking**. Specifically:

1. **Prompt Design:** If your application were to pass scraped content to an LLM for enrichment, what would those prompts look like? Include at least 2–3 example prompts in your codebase or documentation that demonstrate how you would instruct an AI to transform raw scraped data into structured knowledge base fields. These can be in a dedicated `/prompts` directory, in your README, or as constants in your code.
2. **Data Quality Thinking:** How would you handle incomplete data? What fallback strategies would you use when a website doesn't have certain information? Document your approach.
3. **Knowledge Enrichment Ideas:** Beyond what's on the website itself, what other data sources or enrichment strategies could improve a knowledge base? Include your ideas in your documentation.

Questions (Required)

Include a text file with answers to the following:

1. What approach did you take to scraping and structuring the knowledge base data?
2. What information beyond our current baseline did you choose to include, and why?
3. How would your knowledge base design improve the outputs of MoSocial, MoMail, and MoBlogs specifically?
4. What would you improve or change about MoKnowledge if you had more time?
5. What was the most challenging part of this assignment?

Submission Requirements

Submit your project as a link to a GitHub repository with:

1. **Complete source code** with organized file structure
2. **README** explaining:
    - What MoKnowledge does
    - How to set it up and run it
    - Key features and functionality
    - Your approach to scraping and data extraction
    - Your knowledge base schema design
    - Example prompts for AI enrichment
    - Any assumptions or limitations
3. **Screenshots** showing the app in action
4. **JSON output file** containing at least one complete knowledge base example from your scraper
5. **Answers file** with responses to the required questions

Bonus Challenge

Design a Supabase database structure for the knowledge base system. Include:

- Table schemas with column types

- Relationships between tables
- Row Level Security (RLS) policy considerations
- How the schema would support multiple companies and versioning of knowledge base entries

Evaluation Criteria

Area	What We're Evaluating
Scraping Quality	Can you extract meaningful, structured data from a real website?
Knowledge Design	Does your knowledge base structure go beyond the baseline? Is it thoughtful and comprehensive?
Prompting Skills	Do your example prompts demonstrate an understanding of how to get high-quality outputs from an LLM?
Technical Execution	Is the code clean, typed, well-organized, and functional?
Product Thinking	Does your solution demonstrate an understanding of how knowledge powers content generation across multiple applications?
UI/UX	Is the interface intuitive and professional?

Good luck. We're excited to see what you build.